

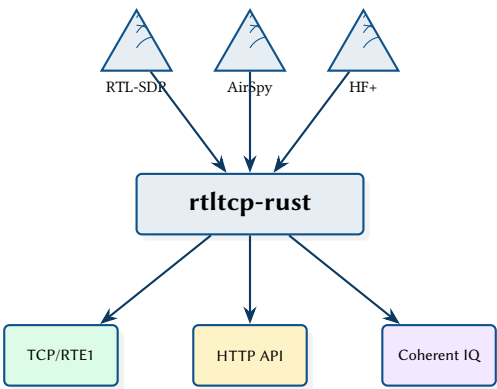
rtltcp-rust: A Multi-SDR Streaming Server with Extended Protocol and Coherent Calibration

Architecture, Protocol Design, and DSP Pipeline

Simon Knight

March 2026 • Version 0.1.0

Last updated: March 11, 2026



Abstract. Standard `rtl_tcp` servers stream unsigned 8-bit IQ samples from a single RTL-SDR device, discarding the higher-resolution native formats of modern SDR hardware. This report presents `rtltcp-rust`, a Rust reimplementation that supports three SDR families (RTL-SDR, AirSpy, AirSpy HF+), preserves native sample depth through a backward-compatible extended protocol (RTE1), and adds adaptive compression, UDP/FEC transport, an HTTP REST API, and a terminal UI. A coherent multi-device mode implements the HeIMDALL calibration FSM for KrakenSDR-class passive radar, producing synchronised multi-channel IQ frames over TCP. The server targets Raspberry Pi deployment via Docker-based cross-compilation.

Version	Date	Changes
0.1.0	March 2026	Initial architecture report

Contents

List of Design Decisions & Rules	2
1 Introduction and Motivation	3
2 System Architecture	3
2.1 Data Pipeline	3
2.2 Device Abstraction	4
2.3 Server State Model	4
3 Extended Protocol (RTE1)	4
3.1 Negotiation	4
3.2 Extended Handshake	5
3.3 Typed Framing	5
3.4 Adaptive Compression	6
3.5 UDP Transport	6
3.6 Extended Commands	6
3.7 Legacy Conversion	7
4 Coherent Multi-Device Mode	7
4.1 Calibration State Machine	7
4.2 DSP Algorithms.	8
4.3 IQ Frame Output	8
5 HTTP API and Terminal UI	9
5.1 REST API.	9
5.2 Terminal UI	9
6 macOS Bridge (mac_tunnel)	9
6.1 Encoding Decode Paths	10
6.2 UDP Mode	10
7 Cross-Compilation and Deployment	11
8 Test Infrastructure	11
8.1 Unit Tests.	11
8.2 Integration Tests	11
9 Design Summary	12

SDR	Software-Defined Radio	3
IQ	In-phase / Quadrature	3
CPI	Coherent Processing Interval	
FSM	Finite State Machine	3
TUI	Terminal User Interface	3
API	Application Programming Interface	3
LE	Little-Endian	5
BE	Big-Endian	5
PPM	Parts Per Million	
UDP	User Datagram Protocol	6
TCP	Transmission Control Protocol	3

List of Design Decisions & Rules

1	Two-Stage Channel Architecture	3
1	Design Rule: Zero-Cost Backward Compatibility	5
2	Big-Endian Control Fields, Little-Endian Sample Data	5
3	Per-Frame Encoding Selection	6
2	Design Rule: Wire Compatibility with HeIMDALL	8
4	C Library stderr Capture	9
5	Delta+LZ4 Keyframe State Machine	10
6	Library Target for Testing	11

1 Introduction and Motivation

The `rtl_tcp` protocol [1] has become the de facto standard for streaming In-phase / Quadrature (IQ) samples from USB Software-Defined Radio (SDR) receivers to remote clients over TCP. Its simplicity—a 12-byte handshake followed by a raw byte stream—has led to wide adoption in applications such as SDR#, GQRX, and CubicSDR.

However, the original protocol has three fundamental limitations:

1. **Single device, single format.** Only RTL-SDR hardware is supported, locked to unsigned 8-bit IQ samples. Modern devices like the AirSpy (signed 16-bit) and AirSpy HF+ (32-bit float) have their samples crushed to 8-bit when served over `rtl_tcp`.
2. **No metadata channel.** Clients cannot discover the server's sample format, sample rate, or frequency without out-of-band coordination.
3. **No multi-device coherence.** Passive radar and direction finding require phase-synchronised multi-channel IQ streams, which the protocol cannot express.

`rtltcp-rust` addresses all three limitations through an extended protocol (RTE1) that negotiates native sample depth at connection time, a broadcast-based fan-out architecture that supports multiple concurrent clients per device, and a coherent multi-device mode that implements the HeIMDALL calibration Finite State Machine (FSM) [2] for KrakenSDR [3] compatibility.

2 System Architecture

The server is a single Rust binary built on the Tokio async runtime [4]. It simultaneously supports a Transmission Control Protocol (TCP) streaming interface per device, an HTTP REST Application Programming Interface (API), an optional Terminal User Interface (TUI), and coherent multi-channel output.

2.1 Data Pipeline

The core data path is a two-stage channel architecture that bridges synchronous C driver callbacks into the async Tokio world, then fans out to an arbitrary number of consumers.

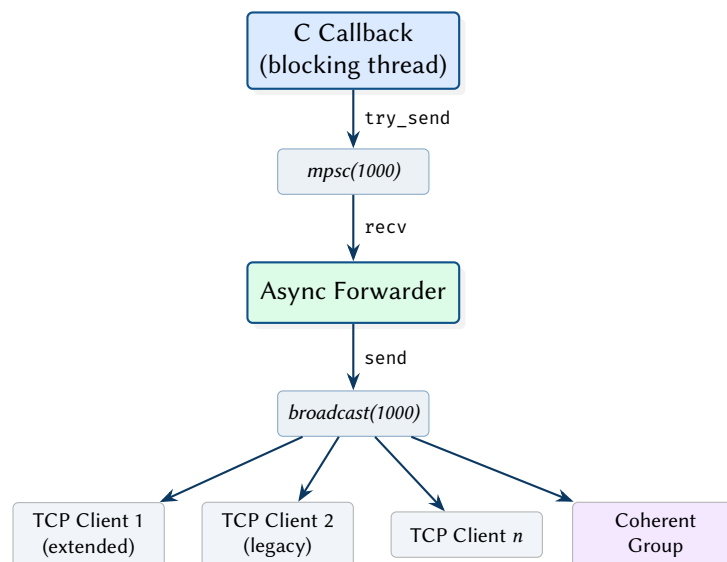


Figure 1. Two-stage data pipeline: device callback to broadcast fan-out.

Design Decision 1: Two-Stage Channel Architecture

The C driver libraries (`librtlsdr`, `libairspyhf`, `libairspy`) deliver samples via blocking callbacks on threads they control. An `mpsc` channel bridges these into the Tokio async world (`try_send` is non-blocking, safe from C callbacks). The `broadcast` channel then provides $O(1)$ fan-out with per-subscriber lag detection—lagged clients receive `RecvError::Lagged(n)` rather than causing backpressure that would block faster consumers.

Insight:
The 100 ms negotiation timeout is the key enabler for backward compatibility: standard clients never speak first, so they naturally fall through to legacy mode with zero code changes.

2.2 Device Abstraction

All SDR hardware is accessed through the SDRDevice trait:

```
#[async_trait]
pub trait SDRDevice: Send + Sync {
    fn set_frequency(&self, freq: u32) -> Result<()>;
    fn set_sample_rate(&self, rate: u32) -> Result<()>;
    fn set_gain(&self, gain: i32) -> Result<()>;
    fn run(&self, tx: mpsc::Sender<SampleBuffer>) -> Result<()>;
    fn stop(&self) -> Result<()>;
    fn sample_format(&self) -> SampleFormat;
    fn capabilities(&self) -> Result<DeviceCapabilities>;
    // ... default no-ops for bias_tee, freq_correction
}
```

Drivers are stored as Arc<dyn SDRDevice> trait objects. The run() method blocks on the C library's async read loop (e.g. rtl_sdr_read_async), so it is always spawned via tokio::task::spawn_blocking().

Table 1. Supported SDR hardware and native sample formats.

Device	Driver	Bits	Format	FFI
RTL-SDR	rtlsdr.rs	8	U8	rtlsdr_sys
AirSpy	airspy.rs	16	I16	Manual #[link]
AirSpy HF+	airspyhf.rs	32	F32	Manual #[link]

2.3 Server State Model

Each device server maintains per-device state wrapped in Arc for shared access across Tokio tasks:

```
struct DeviceServerState {
    device: Arc<dyn SDRDevice>,
    data_tx: broadcast::Sender<SampleBuffer>,
    current_frequency: Arc<Mutex<u32>>,
    current_sample_rate: Arc<Mutex<u32>>,
    current_gain: Arc<Mutex<i32>>,
    bytes_sent: Arc<AtomicU64>,
    // ...
}
```

A TaskManager (HashMap<String, (JoinHandle, CancellationTokentoken)>) controls device lifecycle, allowing the TUI to start and stop individual devices at runtime.

3 Extended Protocol (RTE1)

3.1 Negotiation

The protocol exploits a behavioural asymmetry in standard rtl_tcp clients: they connect and wait for the server's handshake without sending any data first. Extended clients identify themselves by sending the 4-byte magic "RTE1" before the server's 100 ms timeout expires.

Insight:
The AirSpy and AirSpy HF+ drivers use full manual FFI declarations rather than a -sys crate because no maintained Rust bindings exist for these libraries. The build.rs script uses pkg-config for linker path discovery, falling back to sysroot-relative paths for cross-compilation.

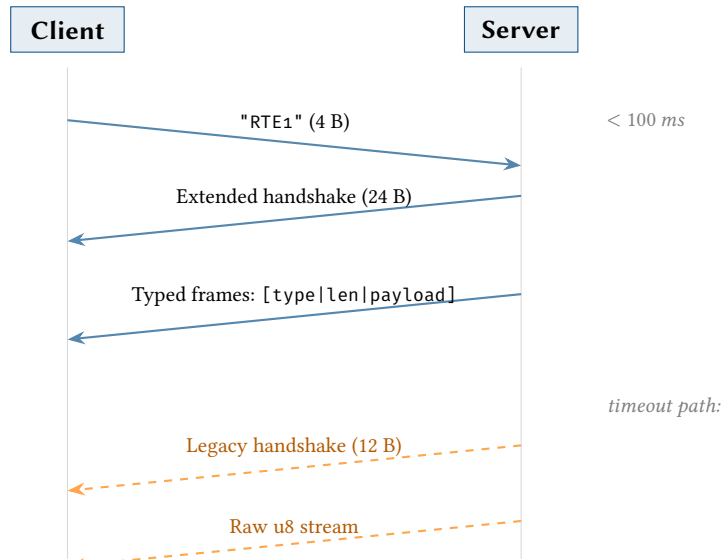


Figure 2. Protocol negotiation: extended vs. legacy path.

: Zero-Cost Backward Compatibility

The extended protocol must not require any changes to existing `rtl_tcp` clients. A timeout-based detection mechanism ensures that legacy clients receive the standard 12-byte RTL0 handshake and raw u8 stream with no modifications.

3.2 Extended Handshake

The 24-byte extended handshake advertises the device's native sample format, current tuning parameters, and server capabilities:

Table 2. Extended handshake wire format (24 bytes, server → client).

Offset	Size	Field	Encoding	Description
0	4	magic	ASCII	"RTE1" echo
4	1	sample_format	u8	0=u8, 1=i16le, 2=f32le
5	1	bits_per_sample	u8	8, 16, or 32
6	1	capability_flags	u8	Bit 0: stats frames
7	1	reserved	zero	—
8	4	sample_rate	u32 Big-Endian (BE)	Sample rate (Hz)
12	4	center_freq	u32 BE	Center frequency (Hz)
16	4	device_serial	u32 BE	Lower 32 bits of serial
20	4	reserved	zero	—

Design Decision 2: Big-Endian Control Fields, Little-Endian Sample Data

The handshake and command fields use big-endian (network byte order) for consistency with the original `rtl_tcp` protocol. Sample data uses little-endian because both ARM (Raspberry Pi) and x86 targets are natively Little-Endian (LE), avoiding byte-swap costs on the critical data path.

3.3 Typed Framing

In extended mode, each transmission unit is a typed frame:

Table 3. Frame header format (5 bytes).

Offset	Size	Field	Encoding
0	1	frame_type	u8
1	4	payload_length	u32 LE

Frame types:

- `0x00` (**Data**) — raw [IQ](#) in negotiated format
- `0x01` (**Stats**) — JSON telemetry (~1 Hz)
- `0x02` (**DataEncoded**) — compressed/encoded data with 8-byte sub-header

3.4 Adaptive Compression

For each sample buffer, the server evaluates all enabled encodings and selects the smallest representation that beats the baseline by at least `min_gain_bytes` (default 64):

1. **Raw / quantise-to-u8** — baseline size
2. **LZ4** [5] — block compression
3. **Delta + LZ4** — wrapping-subtraction delta coding followed by LZ4; first frame is a keyframe

Design Decision 3: Per-Frame Encoding Selection

Rather than committing to a single encoding for the session, the server picks the best encoding per frame. This adapts naturally to varying signal conditions: delta+LZ4 compresses well for narrowband signals (high sample-to-sample correlation) while raw may win for wideband noise where compression overhead exceeds savings.

Encoded frames carry an 8-byte sub-header:

Table 4. Encoded payload sub-header (8 bytes).

Offset	Size	Field	Encoding
0	1	encoding_id	u8 (1=LZ4, 2=quant, 3=delta+LZ4)
1	1	decoded_format	u8 (SampleFormat)
2	1	flags	u8 (bit 0 = KEYFRAME)
3	1	reserved	zero
4	4	decoded_len	u32 LE (decompressed byte size)

3.5 UDP Transport

Two User Datagram Protocol ([UDP](#)) datagram formats complement the primary [TCP](#) stream for low-latency or lossy-link scenarios:

RTU1 (no FEC) 20-byte header: magic "RTU1" + stream ID + sequence number + payload. Payload chunked at 1200 bytes (MTU-safe). Socket send buffer expanded to 1 MB via `socket2`.

RTU2 (RaptorQ FEC) 32-byte header with RaptorQ [6, 7] Object Transmission Information. 64 KiB blocks with 8 repair packets per block, enabling reconstruction from any subset of $k + 8$ received packets.

3.6 Extended Commands

Standard `rtl_tcp` commands (5-byte format) work in both modes. Extended-mode adds:

Insight:
UDP transport always converts to u8 IQ regardless of the device's native format. This simplifies the receiver and keeps datagram sizes small—a single f32 sample pair is 8 bytes vs. 2 bytes in u8.

Table 5. Extended command set (client → server).

Byte	Command	Parameter
0x01	Set frequency	Hz (u32 BE)
0x02	Set sample rate	Hz (u32 BE)
0x04	Set gain	Device-specific
0x80	SET_LZ4	0=off, 1=on
0x81	SET_QUANTIZE_U8	0=off, 1=on
0x82	SET_MIN_GAIN	Minimum byte savings
0x83	SET_UDP_PORT	0=off, 1–65535=port
0x84	SET_UDP_FEC	0=off, 1=on
0x85	SET_DELTA_LZ4	0=off, 1=on

3.7 Legacy Conversion

When a standard client connects, higher-resolution samples are converted server-side to unsigned 8-bit:

$$i16 \rightarrow u8 : ((sample \gg 8) \text{ as } u8) \oplus 0x80 \quad (1)$$

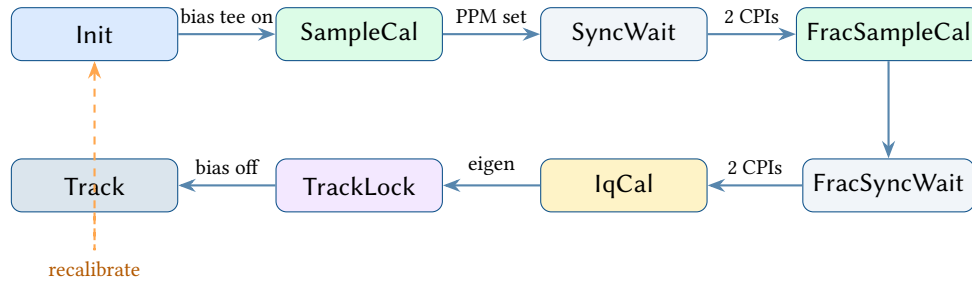
$$f32 \rightarrow u8 : \text{clamp}(sample \times 127.5 + 127.5, 0, 255) \quad (2)$$

Both are standard offset-binary mappings that preserve the DC bias point at sample value 127.

4 Coherent Multi-Device Mode

The coherent mode groups N RTL-SDR devices and runs an 8-state calibration FSM to synchronise their local oscillators and correct per-channel amplitude/phase offsets. The output is a multi-channel IQ frame compatible with the KrakenSDR passive radar receiver [3].

4.1 Calibration State Machine

**Figure 3.** Calibration FSM state transitions.**Table 6.** Calibration FSM actions per state.

State	Action
Init	Enable bias tee on reference device (drives noise source)
SampleCal	FFT cross-correlation per channel → integer sample delays → PPM corrections
SyncWait	Discard 2 CPIs while hardware settles
FracSampleCal	Block-FFT spectral phase slope → fractional delay refinement
FracSyncWait	Discard 2 CPIs
IqCal	Spatial correlation matrix → eigendecomposition → amplitude/phase corrections
TrackLock	Apply corrections in-place; disable noise source
Track	Steady-state: apply corrections, build IqHeader, output frames

4.2 DSP Algorithms

The calibration pipeline implements four core algorithms, all ported from HeIMDALL's `delay_sync.py` [2].

4.2.1 Integer Delay via Cross-Correlation

Given reference signal $x_{\text{ref}}[n]$ and channel signal $x_{\text{ch}}[n]$, the integer sample delay is estimated by:

$$\hat{\tau} = \arg \max_k |\mathcal{F}^{-1}\{X_{\text{ref}}^*(f) \cdot X_{\text{ch}}(f)\}[k]| \quad (3)$$

where $X(f) = \mathcal{F}\{x[n]\}$ is the DFT, computed after zero-padding both signals to $2N$ (next power of two) to avoid circular aliasing [8]. The circular index is unwrapped to a signed lag.

4.2.2 Fractional Delay via Spectral Phase Slope

For sub-sample precision, the signal is divided into blocks of size B (default 1024). Per block, the transfer function $H(f) = X_{\text{ch}}(f)/X_{\text{ref}}(f)$ is computed (bins below -60 dB magnitude are masked). The unwrapped phase $\angle H(f)$ is fit via least-squares to extract the slope [9]:

$$\hat{\tau}_{\text{frac}} = -\frac{\text{slope} \cdot B}{2\pi} \quad (4)$$

The per-block estimates are averaged across all valid blocks in the [CPI](#).

4.2.3 Spatial Correlation Matrix

The $N \times N$ Hermitian correlation matrix is:

$$\mathbf{R}_{xx} = \frac{1}{M} \mathbf{X} \mathbf{X}^H \quad (5)$$

where $\mathbf{X} \in \mathbb{C}^{N \times M}$ contains N channels of M samples each. The implementation uses `nalgebra` with `DMatrix<Complex<f64>>`.

4.2.4 IQ Calibration via Eigendecomposition

The dominant eigenvector of $\mathbf{R}_{xx}$ encodes the per-channel amplitude and phase offsets relative to the reference channel. For each channel i :

$$a_i = |v_i|/|v_0| \quad (\text{amplitude correction}) \quad (6)$$

$$\phi_i = \angle v_i - \angle v_0 \quad (\text{phase correction}) \quad (7)$$

where $\mathbf{v}$ is the eigenvector corresponding to the largest eigenvalue. Corrections are applied in-place: $s[n] \leftarrow s[n] \cdot a_i \cdot e^{j\phi_i}$.

4.3 IQ Frame Output

The coherent output uses a 1024-byte binary header matching HeIMDALL's `iq_header.py` format, followed by per-channel complex-float data:

$$\text{Frame} = \underbrace{[\text{IqHeader}]}_{1024 \text{ B}} \underbrace{[\text{ch}_0][\text{ch}_1] \dots [\text{ch}_{N-1}]}_{N \times \text{CPI} \times 8 \text{ B}} \quad (8)$$

For a 5-channel KrakenSDR with $\text{CPI} = 262,144$, each frame is $1024 + 5 \times 262,144 \times 8 \approx 10$ MB.

: Wire Compatibility with HeIMDALL

The `IqHeader` must be byte-for-byte compatible with `iq_header.py` to allow `krakenSDR_receiver.py` to consume the stream in Ethernet mode without modification. All fields use little-endian encoding to match Python's `struct.pack` default on ARM/x86.

Key header fields:

Table 7. Selected IqHeader fields (1024 bytes total).

Offset	Type	Field	Notes
0	u32	sync_word	0x2BF7B95A
4	u32	frame_type	0=Data, 1=Dummy, 3=Cal
28	u32	active_ant_chs	Number of channels
60	u32	cpi_length	Samples per channel
228	u32	delay_sync_flag	1 when sample-aligned
232	u32	iq_sync_flag	1 when IQ-calibrated
236	u32	sync_state	CalState as u32
240	u32	noise_source	1 when bias tee active

5 HTTP API and Terminal UI

5.1 REST API

The HTTP [API](#) is built on Axum [10] and serves JSON responses. All state is accessed through an `Arc<ApiState>` injected via Axum’s State extractor.

Table 8. HTTP API endpoints.

Method	Path	Description
GET	/api/v1/health	Liveness check
GET	/api/v1/server	Version, uptime, device count
GET	/api/v1/devices	All devices with status
GET	/api/v1/devices/{n}	Single device (404 if absent)
GET	/api/v1/devices/{n}/capabilities	Hardware caps (503 if loading)
GET	/api/v1/system	CPU, memory, temperature
GET	/api/v1/coherent	List coherent groups
POST	/api/v1/coherent	Create coherent group
DELETE	/api/v1/coherent/{n}	Destroy group
POST	/api/v1/coherent/{n}/recalibrate	Restart FSM

5.2 Terminal UI

The [TUI](#) [11] provides two tabs—**Devices** (real-time status, inline parameter input) and **Logs** (scrolling output). Device commands flow over a `broadcast::channel` so all device server tasks see them simultaneously; each ignores commands not addressed to it.

Design Decision 4: C Library `stderr` Capture

C driver libraries print USB discovery messages directly to file descriptor 2. In raw terminal mode these lack carriage returns, causing “staircase” rendering. The server captures `stderr` around driver calls via `POSIX pipe()/dup2()`, drains the pipe, and routes captured output through the TUI log channel.

6 macOS Bridge (`mac_tunnel`)

Many popular macOS [SDR](#) applications—CubicSDR, GQRX, SDR# (via Mono)—speak only the legacy RTL0 `rtl_tcp` protocol: a 12-byte handshake followed by an unsigned 8-bit [IQ](#) stream. The `mac_tunnel` binary acts as a transparent protocol bridge: it connects upstream to an `rtltcp-rust` server using the extended RTE1 protocol (receiving native sample depth and adaptive compression), then re-serves the data locally as a standard RTL0 server that any legacy application can connect to without modification.

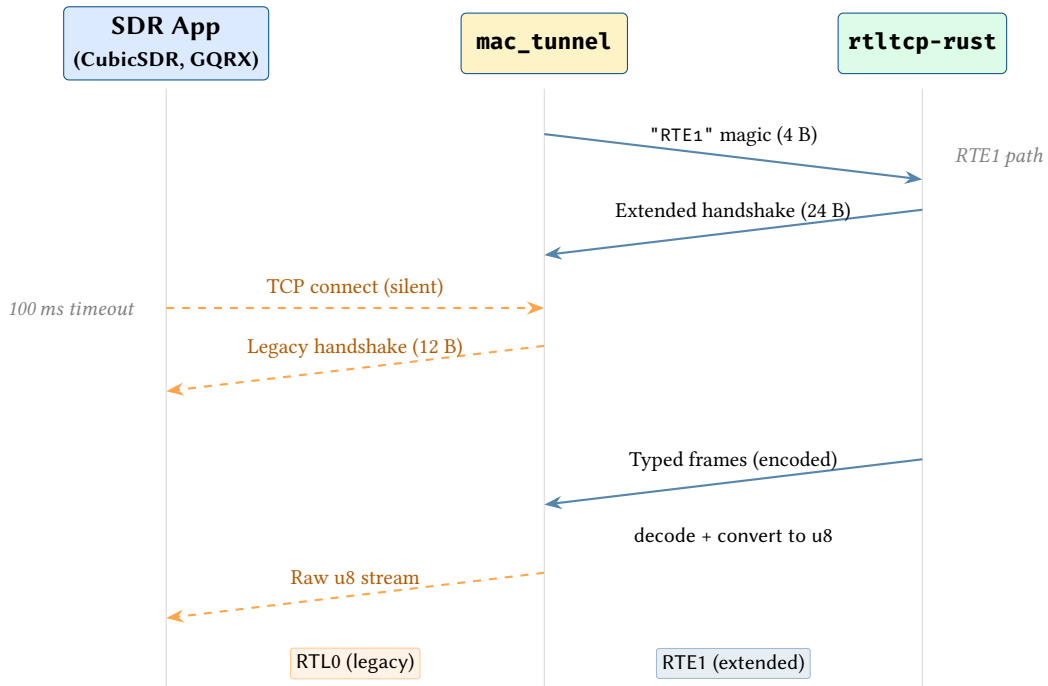


Figure 4. Protocol bridge: `mac_tunnel` translates between RTE1 and RTL0.

6.1 Encoding Decode Paths

When the upstream server sends encoded frames (`0x02 DataEncoded`), the tunnel must decode them before converting to raw u8 for the legacy client. The decode path depends on the `encoding_id` in the 8-byte sub-header:

Table 9. Tunnel decode paths by encoding type.

ID	Encoding	Decode Path
1	LZ4	Decompress via <code>lz4_flex</code> ; convert samples to u8
2	Quantize-u8	Pass through (payload is already unsigned 8-bit)
3	Delta+LZ4	Decompress via <code>lz4_flex</code> ; apply delta to reference frame; update reference; convert to u8

Design Decision 5: Delta+LZ4 Keyframe State Machine

The delta+LZ4 encoding requires the tunnel to maintain a reference frame for each active stream. The `flags` byte in the encoded payload sub-header (bit 0 = KEYFRAME) drives a two-state machine:

- **Keyframe received** (`flags & 0x01`): decompress the LZ4 payload and store the result as the new reference frame. Output this frame directly.
- **Delta frame** (no keyframe flag): decompress the LZ4 payload to obtain the delta buffer, then apply wrapping addition to the current reference frame: `ref[i] += delta[i]`. Output the updated reference frame.

The client *must* track reference frame state; a missed keyframe causes all subsequent delta frames to produce corrupt output until the next keyframe arrives.

6.2 UDP Mode

In addition to TCP, `mac_tunnel` can receive samples over UDP using the RTU1 and RTU2 datagram formats. A playout buffer reorders and dejitters incoming datagrams before feeding the decode pipeline.

RTU1 (no FEC) Simple [UDP](#) datagrams with 20-byte header: magic "RTU1" + stream ID + sequence number + payload. Datagrams are inserted into the playout buffer keyed by sequence number. Out-of-order packets are reordered; gaps beyond the playout window are treated as losses and zero-filled.

RTU2 (RaptorQ FEC) 32-byte header carrying RaptorQ [6, 7] Object Transmission Information. The playout buffer collects encoding symbols per source block and attempts reconstruction once k or more symbols arrive. With 8 repair symbols per block, the receiver tolerates up to 8 lost datagrams per 64 KiB block.

The playout buffer is parameterised by a target depth (default 20 ms), expressed as a number of sequence slots. Packets arriving later than the playout deadline are silently discarded. The buffer emits samples in strict sequence order, bridging the unreliable [UDP](#) transport into the ordered byte stream expected by legacy RTLO clients.

7 Cross-Compilation and Deployment

The server targets Raspberry Pi via the `cross` tool with custom Docker images that cross-compile `librtlsdr`, `libairspyhf`, and `libairspy` from source for the target architecture.

Table 10. Cross-compilation targets.

Target triple	Pi model	Docker image
aarch64-unknown-linux-gnu	Pi 3/4/5 (64-bit)	rtltcp-cross:aarch64
armv7-unknown-linux-gnueabihf	Pi 2/3 (32-bit)	rtltcp-cross:armv7
arm-unknown-linux-gnueabihf	Pi Zero/1	rtltcp-cross:arm

The `deploy-to-pi.sh` script handles the full pipeline: cross-build, optional strip, scp to Pi, and USBFS buffer configuration (256 MB recommended for [SDR](#) streaming).

All DSP dependencies (`rustfft`, `ndarray`, `nalgebra`, `num-complex`) are pure Rust and cross-compile without issue.

8 Test Infrastructure

8.1 Unit Tests

The protocol module (`protocol.rs`) contains 30+ inline tests covering:

- `SampleBuffer` serialisation/deserialisation roundtrips for all three formats
- Misalignment rejection (odd bytes for I16, non-multiple-of-4 for F32)
- Handshake build/parse roundtrips (both extended and legacy)
- Frame header encoding/decoding
- Encoded payload roundtrips (LZ4, quantise, delta+LZ4)
- UDP RTU1 and RTU2 datagram build/parse with bad-magic and too-short rejection

The calibration module (`calibration.rs`) tests all DSP functions with synthetic signals, verifying cross-correlation delay detection, fractional delay estimation accuracy (< 0.05 samples), spatial correlation matrix properties, and IQ calibration recovery of known amplitude/phase offsets.

8.2 Integration Tests

Integration tests in `tests/` use a `MockDevice` implementing `SDRDevice` to test the coherent [FSM](#) without hardware. The end-to-end test spawns the full `run_coherent_group()` task, feeds synthesised tone data via broadcast channels, connects a TCP client, reads IQ frames, and asserts the FSM reaches the Track state.

Design Decision 6: Library Target for Testing

The crate is primarily a binary, but `lib.rs` re-exports the modules needed by integration tests. This allows `cargo test` to compile tests against the library target without coupling to `main.rs` specifics or requiring a running server.

Insight: UDP mode is particularly useful for macOS clients connected over Wi-Fi, where the combination of RTU2 FEC and the playout buffer masks transient packet loss that would otherwise cause TCP retransmission stalls.

9 Design Summary

Table 11. Key design decisions and their rationale.

Decision	Rationale
Broadcast channel fan-out	$O(1)$ fan-out with per-subscriber lag detection; lagged clients get <code>RecvError::Lagged</code> rather than blocking faster consumers
Two-stage mpsc $\rightarrow$ broadcast	C callbacks cannot be async; mpsc bridges the synchronous C world into Tokio
100 ms timeout negotiation	Standard clients never speak first, so timeout naturally selects legacy mode
Per-frame encoding selection	Adapts to varying signal conditions without session-level commitment
LE sample data, BE control fields	LE avoids byte-swap on ARM/x86 data path; BE matches original protocol convention
stderr capture via dup2	Prevents C library writes from corrupting TUI raw terminal rendering
Manual FFI for AirSpy	No maintained Rust bindings exist; manual declarations avoid unmaintained dependency

References

- [1] S. Markgraf, D. Stolnikov, and Hoernchen, “librtlsdr: Turns your Realtek RTL2832 based DVB dongle into a SDR receiver,” in *osmocom project*, 2012. [Online]. Available: <https://github.com/osmocom/rtl-sdr>
- [2] KrakenRF, *HeIMDALL DAQ Firmware*, 2023. [Online]. Available: https://github.com/krakenrf/heimdall_daq_fw
- [3] C. Laufer et al., “KrakenSDR: Five channel coherent RTL-SDR receiver,” in *Crowd Supply*, 2022. [Online]. Available: <https://www.crowdsupply.com/krakenrf/krakensdr>
- [4] Tokio Contributors, *Tokio: An asynchronous runtime for Rust*, 2024. [Online]. Available: <https://tokio.rs>
- [5] Y. Collet, “LZ4: Extremely fast compression,” 2011. [Online]. Available: <https://lz4.org>
- [6] A. Shokrollahi, “Raptor codes,” in *IEEE Transactions on Information Theory*, vol. 52, 2006, pp. 2551–2567.
- [7] M. Luby, A. Shokrollahi, M. Watson, T. Stockhammer, and L. Minder, “RaptorQ forward error correction scheme for object delivery,” IETF, RFC 6330, 2011.
- [8] G. C. Carter, “Coherence and time delay estimation,” *Proceedings of the IEEE*, vol. 75, no. 2, pp. 236–255, 1987.
- [9] R. C. Williamson, “Delay estimation using narrow-band processes,” *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 29, no. 3, pp. 577–581, 1981.
- [10] Tokio Contributors, *Axum: Ergonomic and modular web framework built with Tokio*, 2024. [Online]. Available: <https://github.com/tokio-rs/axum>
- [11] Ratatui Contributors, *Ratatui: A Rust library for cooking up terminal user interfaces*, 2024. [Online]. Available: <https://ratatui.rs>